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jQuery

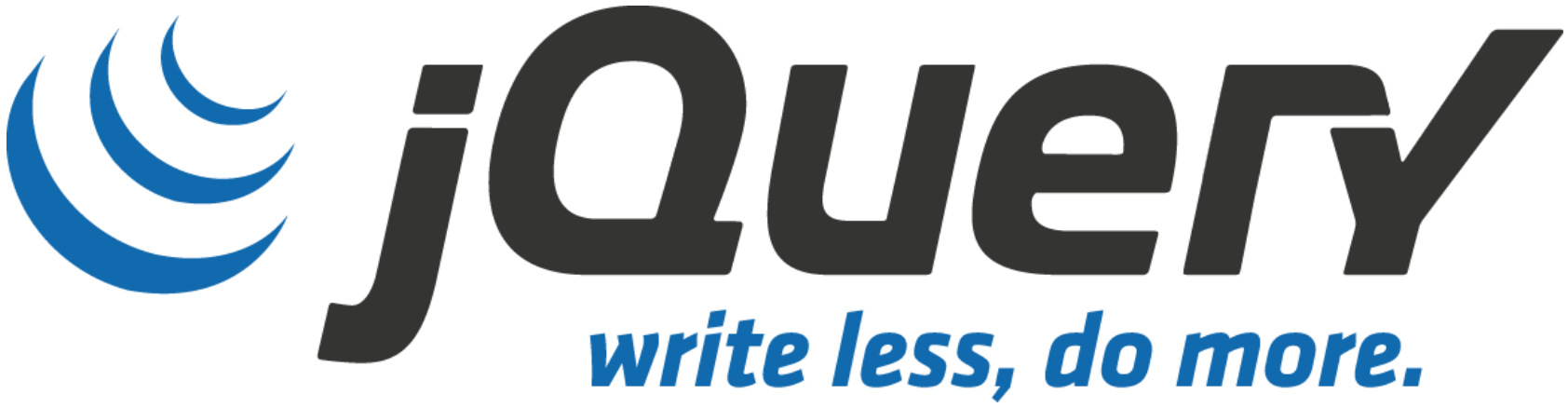
CIS 2102 - Review

DCISM
DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

Topics

1. Introduction
2. Basics
3. DOM Manipulation
4. Get/Set Attributes
5. Events
6. Effects
7. Ajax

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Introduction

- jQuery is an open source JavaScript library that simplifies the interactions between an HTML/CSS document, or more precisely the Document Object Model (DOM), and JavaScript.
- jQuery offers a simple way to achieve a variety of common JavaScript tasks quickly and consistently, across all major browsers.

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- It simplifies HTML document traversing and manipulation, event handling, DOM animations, Ajax interactions, and cross-browser JavaScript development.
- It lets you find HTML elements using CSS-style selectors and do something with the element using jQuery methods.

How to find elements using CSS-style Selectors

- `jQuery()` function lets you find elements in the HTML page. It creates an object which holds references to those HTML elements like using the `document.getElementById(), document.querySelector()` and etc..
- `$()` is often used as a shortcut to save time and space rather than typing `jQuery()`.

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$('li.link')

- This `jQuery()/$()` function accepts 1 parameter which is a CSS-style selector.
- This finds all `<li>` elements with a class of link and makes a reference of it.

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Comparison with JavaScript DOM queries

- Similar task with js DOM queries albeit a simpler syntax
- References of the jQuery object can be also be stored in a variable like in js
- It can be used to manipulate HTML elements using jQuery method

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How to use jQuery

- Download the jQuery file at https://jquery.com/download/ and put it in your project folder then link it in your HTML through <script> tag.
- Use CDN (Content Delivery/Distribution Network) and use it as a source in your <script> tag

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CDNs

- Google CDN

```
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquer
y.min.js"></script>
```

- Microsoft CDN

```
<script src="https://ajax.aspnetcdn.com/ajax/jQuery/jquery-
3.4.1.min.js"></script>
```

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jQuery Selectors

- `$("*")` – all elements (multiple)
- `$("element/tag")` – all elements/tag with that element/tag name (multiple)
- `$("#id")` – element with specified id (single)
- `$("input.class")` – element with specified class (multiple)
- `$("element:first")` – first filtered element with that element/tag name (single)

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DOM Manipulation



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Functions

- `.ready()` – function that checks if the page is ready

```
$(document).ready(function(){
      //code goes here for when the page is ready
});
```

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```
…
<ul>
        <li id="one"><em>fresh</em> figs</li>
        <li id="two">pine nuts</li>
        <li id="three">honey</li>
        <li id="four">balsamic vinegar</li>
</ul>
<p id="p1">TEXT1</p>
<p id="p2">TEXT2</p>
<p class="pClass">REPLACE THIS TEXT</p>
<p id="removeMe">Remove Me</p>
…
```

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- `.html()` – used to retrieve/update information from a jQuery selection, it retrieves only the HTML inside the first element

```
//retrieve
console.log($('li:first').html());
//update
$('p#p2').html(function(){
      return "<h1>BLABLA</h1>";
});
```

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- `.text()` – used to retrieve/ update text from a jQuery selection

```
//retrieve
console.log($('li:first').text());
//update
$('p#p2').text(function(){
      return "BLABLA";
});
```

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- `.replaceWith()` – replaces every element in a matched set with new content.

```
$('p.pClass')[0].replaceWith("BLABLA3");
```

- `.remove()` – removes all elements in the matched set of selectors

```
$('p#removeMe').remove();
```

- `.before()` – inserts content before the selected element

```
$('ul').before('<p class="before">BEFORE</p>');
```

- `.after()` – inserts content after selected element

```
$('ul').after('<p class="after">AFTER</p>');
```

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- `.prepend()` – inserts content inside the selected element after the opening tag

```
$('ul').prepend('<li id="prepend">PREPEND</li>');
```

- `.append()` – inserts content inside the selected element before the closing tag

```
$('ul').append('<li id="append">APPEND</li>');
```

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Traversing the DOM

- Selector is required

Method	Description
.find()	All elements within current selection that match selector
.closest()	Nearest ancestor (parent, siblings, children) that matches selector

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- Selector is optional

Method	Description
.parent()	Direct parent of current selection
.parents()	All parents of current selection
.children()	All children of current selection
.siblings()	All siblings of current selection
.next()	Next sibling of current element
.nextAll()	All subsequent siblings of current element
.prev()	Previous sibling of current element
.prevAll()	All previous siblings of current element

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Get/Set Attributes



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```
<style>
        .inputBG {
          background-color: Red;
          color: yellow;
        }
        .inputHover:hover {
          background-color: gray;
          color: black;
        }
</style>
…
<input id="inputID" class="inputBG inputHover" type="text"
placeholder="Input here" value="This is a text"/>
<button onclick="attributesFunction()">Click Me</button>
…
…
function attributesFunction(){
       var inputField = $("#inputID");
       …
}
…
```

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- `.attr()` – gets/sets attribute and its value. To get the value, specify the attribute name as the only parameter. To update or set an attribute, specify the attribute's name and the value as parameters.

```
console.log(inputField.attr("type"));//get

inputField.attr("type", "password");//set
```

- `.removeAttr()` – To remove an attribute, specify the attribute name as the only parameter.

```
inputField.removeAttr("value");
```

- `.addClass()` – adds a new class to the existing classes. It does not overwrite the existing ones

```
inputField.addClass("inputHover");
```

- `.removeClass()` – removes a specified class from the list of class that exists

```
inputField.removeClass("inputBG");
```

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CSS Manipulation

- **Get**
  - .css() – To get the value, specify the property as the only parameter.

```
var bgColor = inputField.css("background-color");
```

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- **Set**
  - Single property – Specific property and its value are the 2 needed parameters

```
inputField.css("background-color", "blue");
inputField.css("height", "+=80");
```

  - Multiple properties – takes in an object as parameter

```
inputField.css({
        "background-color": "blue",
        "height": "100px",
        "font-size": "30px",
        "color": "pink"
});
```

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Each

- `.each()` – recreates the loop functionality on a selection of elements
- `this/$(this)` – as the .each() function goes through the elements in a selection, you can access the current element by using it

```
$("li").each(function(){
        console.log($(this).html());
        $(this).html("<li>BABA</li>");
});
```

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Events

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- Uses `.on()` to handle all events.

Syntax:

```
$("#btn").on("click", function(){
      console.log("Button is clicked");
});
$("#btn").on("click", buttonClicked);
function buttonClicked(){
      console.log("Button is clicked");
}
```

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- Common events

UI/Element	focus, blur, change
Keyboard	input, keydown, keyup, keypress
Mouse	click, dbclick, mouseup, mousedown, mouseover, mousemove, mouseout, hover
Form	submit, select, change
Document/Page	ready, load, unload
Browser	error, resize, scroll

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- Event Objects – contains methods, properties and information related to the event that occurred.

Properties	Description
type	Type of event, (e.g., click, mouseover)
which	Button or key that was pressed
data	An object extra information passed to the function when the event fi res (See right-hand page for an example)
target	Element that initiated the event
pageX	Mouse position from left edge of viewport
pageY	Mouse position from top of viewport
timeStamp	Number of milliseconds from Jan 1st, 1970, to when the event was triggered (this is known as Unix Time). Does not work in Firefox.

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Methods	Description
.preventDefault()	Prevents the default (e.g., submitting a form reloads the page)
.stopPropagation()	Stops the event bubbling up to ancestors

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Effects

Enhances your web page’s transitions and movements

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- Basic Effects: can take “slow”, “fast” or milliseconds as parameters

Method	Description
.show()	Displays selected elements
.hide()	Hides selected elements
.toggle()	Toggles between showing and hiding selected elements

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- Fading Effects: can take “slow”, “fast” or milliseconds as parameters

Method	Description
.fadeIn()	Fades in selected elements making them opaque
.fadeOut()	Fades out selected elements making them transparent
.fadeTo	Changes opacity of selected elements
.fadeToggle()	Hides or shows selected elements by changing their opacity (the opposite of their current state)

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- Sliding Effects: can take “slow”, “fast” or milliseconds as parameters

Method	Description
.slideUp()	Shows selected elements with a sliding motion
.slideDown()	Hides selected elements with a sliding motion
.slideToggle()	Hides or shows selected elements with a sliding motion (in the opposite direction to its current state)

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- Custom Effects

Method	Description
.delay()	Delays execution. Can take “slow”, “fast” or milliseconds as parameters.
.stop()	Stops an animation if it is currently running
.animate()	Creates custom animations

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- `.animate()` – takes in CSS objects, milliseconds and function as parameters

```
$("#btnEffects").animate({
      "opacity":"0.0",
      "height": "0px",
      "width": "0px"
}, 1000, function(){
      $("#btnEffects").remove();
});
```

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Ajax

(**A**synchronous **J**avaScript **A**nd **X**ML)

Technique for loading data from a server into part of a page without having to refresh the entire page. The data is often sent in a format called JavaScript Object Notation (or JSON).

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- Ajax uses an asynchronous processing model. This means the user can do other things while the web browser is waiting for the data to load, speeding up the user experience.
- Returns either HTML, XML or JSON data.
- XML – (eXtensible Markup Language) looks a lot like HTML but the tag contains different names because they describe the data they contain.
- JSON – (JavaScript Object Notation) uses very similar syntax to object literal notation/JavaScript objects.

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How Ajax works:

1. Event occurs in a web page(load, click, etc.)
2. XMLHttpRequest object is created
3. XMLHttpRequest object sends a request to a server
4. Server processes the request
5. Server sends a response back to the web page
6. Response is read by JavaScript
7. Action is done like updating a web page with response data

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XMLHttpRequest Object

used to exchange data with a server behind the scenes

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- Create an XMLHttpRequest Object:

```
var xhttp = new XMLHttpRequest();
```

Methods	Description
.open()	Accepts these 3 parameters: method: type of request which can either be **GET** or **POST** url: server file/location async: can wither be true or false
.send()	GET: has no parameter
	POST: has a string parameter that is to be sent to the server

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- **GET** is used to request for data from server
  - Simple:

```
xhttp.open("GET", "demo.text", true);
xhttp.send();
```

  - With additional information:

```
xhttp.open("GET", "demo.asp?id=001&name=Doe", true);
xhttp.send();
```

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- **POST** is used to send data to server for storage
  - Simple:

```
xhttp.open("POST", "demo.asp", true);
xhttp.send();
```

  - Specifying data to send to server:

```
xhttp.open("POST", "demo.asp", true);
xhttp.setRequestHeader("Content-type", "application/x-www-form-urlencoded");
xhttp.send("id=001&name=Doe");
```

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- `.onreadystatechange` is where a function is defined when the request receives a response from the server. Use “`this`” to access the properties from the response data.

```
xhttp.onreadystatechange = function() {
      if (this.readyState == 4 && this.status == 200) {
            console.log(this.responseText);
      }
};
```

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Properties	Description
responseText	Text-based data returned
responseXML	XML data returned
readyState	Holds the status of the XMLHttpRequest. 0 - request not initialized 1 - server connection established 2 - request received 3 - processing request 4 - request finished and response is ready
status	Status code. 200 - "OK" 403 - "Forbidden" 404 - "Page not found"
statusText	Status description

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- If the response is a JSON string through `responseText`, then it must be converted to JavaScript object using
  - `var ob = jQuery.parseJSON(this.responseText);`
  - `var ob = JSON.parse(this.responseText);`
- If the response is an XML through `responseXML`, it is better to convert it to an HTML table.

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API

- Application Programming Interface
- A software intermediary that allows two applications to talk to each other
- The API is the middleman between the application and the server

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Sample:

- When you use a mobile application, it connects to the Internet and sends data to a server.
- The server then retrieves that request, interprets it, performs the necessary actions and sends it back to your phone.
- The application then interprets that data and presents you with the information you wanted in a readable way.

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Reference

- Ducket, J. (2014). *JAVASCRIPT & JQUERY Interactive Front-End Web Development.* Indianapolis, Indiana: John Wiley & Sons, Inc.

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